INTRODUCTION:

The LEED Green Building Rating Systems reward project teams with points toward certification
based on a building's environmental and energy performance. While LEED does not assign points for products, the materials and building product that a manufacturer provides have a very important role in achieving LEED certification. The following is an assessment of how Flex Trim Cured Polyurethane flexible trim products can assist project teams with certification through the LEED for New Construction and Major Renovations v.2.2 rating system (LEEDNC).

SUSTAINABLE ATTRIBUTES OF FLEX TRIM CURED POLYURETHANE:

Flex Trim™ Cured Polyurethane flexible trim is "greener" than polyester resin based flexible moulding and is much more durable than traditional wood trim moulding. It can be cut with a saw, caulked, and painted with water-based paints to complement nearly any interior color scheme. The sustainable attributes of Flex Trim can be distinguished from most other flexible trim moulding products by the following:

Zero-VOCs: Flex Trim is 100% solids containing Zero VOCs *(SCAQMD Rule 443.1)* and provide a reduced impact on the environment compared to resin based products such as polyester trim, which can expose building occupants to VOC emissions. Flex Trim products reduce VOCs for factory workers, warehouse and distributorship staff, as well as installers and contractors. Reduced Green House Gas Emissions: The Flex Trim flexible trim does not contribute to green house gas omissions. By comparison, polyester is not 100% solids and it emits styrene as well as other harmful off gassing emissions.

Durability:

A polyurethane based building product also represents a more durable material with a high potential for salvaging during deconstruction for both new uses or reuse.

Indoor Air Quality:

The Flex Trim polyurethane flexible moulding replaces traditional wood molding, which supports rot, pests and fungal growth. Because polyurethane flexible trim does not provide a food source for insects, there is a potential for fewer applications of pesticides with Flex Trim. All of these factors can assist with improved indoor air quality.

Regional Materials:

Flex Trim has facilities in Utah and North Carolina. Building projects within a 500 mile radius of the following zip codes can be considered regional materials: 84601 & 27299

MATERIALS AND RESOURCES (MR)

MR Credit 1.3 Building Reuse: Maintain 50% of Interior Non Structural Elements 1 point

The LEED rating system rewards project teams for utilizing reused building products.

Flex Trim

Flexible trim as part of the casework or a cabinetry system can be included in the overall calculation for determining the percentage of reused materials in order to comply with MR Credit 1.3.

MR Credit 2.1 Construction Waste Management: Divert 50% from Disposal1 point

MR Credit 2.2 Construction Waste Management: Divert 75% from Disposal1 point

LEED requires 50% of non-structural elements reused in order to gain a point in this credit category with a second point available through reuse of 75% of the products. The calculation is completed either by weight or volume of the materials to be reused.

Flex Trim:

Flex Trim flexible trim is strong, reusable and the trim can be redirected to another construction site for installation in another building, donated to a charity, or it can be salvaged during deconstruction and resold.

MR Credit 3.1 Materials Reuse 5% 1 point

MR Credit 3.2 Materials Reuse 10% 1 point

This LEED Credit requires salvaged, refurbished or reused materials to be selected by the project team such that the sum of the materials constitute at least 5%, based on cost, of the total value of the materials on the project. A second point is available for using 10% reused materials on the project.

Flex Trim
The durability and strength of the Flex Trim cured polyurethane flexible trim means project teams can reuse the product in projects where reclaimed building materials are a priority.

MR Credit 5.1 Regional Materials**:** 10% Extracted, Processed & Manufactured Regionally **1 point**

MR Credit 5.2 Regional Materials: 20% Extracted, Processed & Manufactured Regionally **1 point**
The product ingredients must be extracted, processed and manufactured all within a 500 mile radius. The regional materials percentage is determined by the cost of the total material value. If only a fraction of a product or material is extracted/ harvested/ recovered and manufactured locally, then only that percentage (by weight) can contribute to the regional value.

Flex Trim
Flex Trim produces the Flex Trim products in Utah (Zip Code: 84601) and in North Carolina (Zip Code: 27299). Products shipped within this region have less embodied energy than product from outside the region.

INDOOR ENVIRONMENTAL QUALITY (EQ)

EQ Credit 3.2 Construction IAQ Management Plan¬Before Occupancy 1 point

Perform an Indoor Air Quality (IAQ) Management Plan with a building flush-out, before or after occupancy; or conduct air quality testing after construction and prior to occupancy according to EPA standards. The VOC levels cannot exceed 500 micrograms per cubic meter.

Flex Trim
The Zero VOC emissions from the Flex Trim trim can assist building teams with

compliance with this LEED credit by reducing indoor air quality problems during the construction and renovation process for the comfort and well being of installers and occupants.

INNOVATION IN DESIGN (ID)

ID Credit 1 Resource Reuse Program 1 point

The facility manager must implement a program for reintroducing surplus building products in future projects. Products to be reused include caulk and sealants, adhesives, plywood and lumber, paints and lacquers, roofing materials and fasteners. The submittals include a narrative of the program and an inventory list.

Flex Trim

Flex Trim is durable and has similar characteristics of wood trim, the pieces left over after cutting could be entered into the inventory of a resource reuse program. Also if trim is removed during demolition it may be reused on future projects.

ID Credit 2 Post Occupancy Survey 1 point

This innovation credit for determining occupant comfort over time has been accepted by the USGBC in the past. The requirement is that the building team submit a survey showing overall building user satisfaction be measured for thermal comfort, general satisfaction, layout, furnishings, air quality, lighting, acoustic quality and cleanliness. The survey is typically conducted 18 months after occupancy.

Flex Trim

Flex Trim flexible trim contains Zero VOCs and can assist building teams with compliance with this credit category because of its contribution to better indoor air quality.

DISCLAIMER

While the LEED NC credit sections described in this paper suggest how applications of Flex Trim can assist project teams with earning LEED points, the LEED applicant obviously bears the ultimate responsibility for determining the product attributes will assist them with LEED certification. The final decision regarding LEED certification relies heavily on the work and judgment of the LEED Accredited Professional (LEED AP) retained for a specific project and ultimately on the judging panel at USGBC. The architect, designer, contractor, or other member of the building team must document a building's sustainable design, construction, and performance data and make the data available to the LEED AP and the Council. Then the LEED AP can prepare final documentation which is submitted to the USGBC.